Faddev-Popov Method.

1. For gauge field, path-integral is ill-defined

$$Z[J] = \int DA_{gauge}\, e^{iS[A_{gauge}] + i\int d^4x\, A_{gauge} J}$$

Recall fundation of path-integral is that we insert a complete set of eigen. states defined in Hilbert space $\int_{A_x} dA(x)\, |A(x)\rangle\langle A(x)|$ including eigenstates w/ same eigenvalue $A(x)$ (in BRST formalism, physical states are defined uniquely)

These eigenstates have same physical meaning, they are related by gauge transformation. $A^\mu(x) + \partial^\mu \lambda(x)$, they are on same gauge orbit.

$\boxed{\text{Why there is no problem with theory obeying global symmetry? or spontaneously-broken symmetry?}}$

Instead consider generally.

$$\int DA\, e^{iS[A] + i\int d^4x\, AJ}\, B[f[A(x)]]\, \det \frac{\delta f[A(x)]}{\delta \lambda(y)} = Z[J]$$

$\lambda(y)$: gauge transformation parameter.

f: gauge fixing.

(1) first it must be gauge inv.

(2) indp of gauge fixing.

$\boxed{\text{Why it is valid and better then original?}}$

Consider field redefinitein of $A \to A + \partial\Lambda(x) \equiv A_\Lambda$ (not gauge trans.)

$$\underbrace{\int DA_\Lambda\, e^{iS[A_\Lambda]+i\int d^4x\, A_\Lambda J}}_{\text{gauge inv}} B[f[A_\Lambda(x)]]\, \det \frac{\delta f[A_\Lambda(x)]}{\delta \lambda(y)}$$

$$= \int DA\, e^{iS[A]+i\int d^4x\, AJ} B[f[A_\Lambda(x)]]\, \det \frac{\delta f[A_\Lambda(x)]}{\delta \lambda(y)} = Z[B,f,J]$$

Λ- indp on RHS.

$$\int d\Lambda(z)\, \rho(z)\, Z[B,f,J] = \int d\Lambda(z) \overset{\rho(z)}{\int} DA\, e^{iS[A]+i\int d^4x AJ} B[f[A_\Lambda(x)]]\, \det \frac{\delta f[A_\Lambda(x)]}{\delta \lambda(y)}$$

consider $C[B,f,A] = \int d\Lambda(z)\, B[f[A_\Lambda(x)]]\, \det \frac{\delta f[A_\Lambda(x)]}{\delta \lambda(y)}\, \rho(z)$

if C is indp of f and A $\Rightarrow$ we prove $Z[B,f,J] = Z[B,J]$

indp of gauge fixing, and more surprising, it is

$\propto \int DA\, e^{iS+i\int d^4x\, AJ}$, because A_Λ defined like gauge transformed A

$$\int \left.\frac{\delta f[A_{\vec\lambda\cdot\vec\Lambda}(x)]}{\delta\, \vec\lambda\cdot\vec\Lambda(y')}\right|_{\lambda=0} \left.\frac{\delta\, \vec\lambda\cdot\vec\Lambda(y')}{\delta\, \lambda(y)}\right|_{\lambda=0} d^4y' =$$

first act Λ then λ.

$$\left.\frac{\delta f[A_{\vec\lambda\cdot\vec\Lambda}(x)]}{\delta \vec\lambda\cdot\vec\Lambda(y')}\right|_{\lambda=0} = \frac{\delta f[A_\Lambda(x)]}{\delta \Lambda(y')} \quad \text{non-trivially}$$

$$C[B,f,A] = \int d\Lambda(z)\, B[f[A_\Lambda(x)]]\, \det\left(\int \frac{\delta f[A_\Lambda(x)]}{\delta \Lambda(y')} \left.\frac{\delta\, \vec\lambda\cdot\vec\Lambda(y')}{\delta\, \lambda(y)}\right|_{\lambda=0} d^4y'\right) \cdot \rho(z)$$

choose $\rho(z) = \int \frac{1}{\det\left(\frac{\delta\, \vec\lambda\cdot\vec\Lambda(y')}{\delta\, \vec\lambda(z)}\right)} d^4y'$

$$= \int d\Lambda(z)\, \det\left(\frac{f[A_\Lambda]}{\delta\Lambda}\right) B[f[A_\Lambda]] = \int df[A_\Lambda]\, B[f[A_\Lambda]] = C[B] \quad \#$$

$$Z[B,J] = \boxed{\frac{C[B]}{\int d\Lambda\, \rho(z)}} \int DA\, e^{iS[A]+i\int d^4x\, AJ}$$

irrelavent constant.

we know $\int DA\, e^{iS[A]+i\int d^4x\, AJ}$ is divergent cuz $\int DA$ can integrate along gauge orbit with e^{iS} phase not decaying

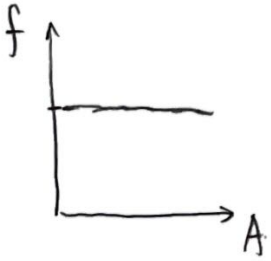
$\Rightarrow$ divergences cancelled by $\int d\Lambda\, \rho(z)$

$B[f[A(x)]] = e^{-\frac{i}{2\xi}\int d^4x\, f^2[A]}$ we choose an arbitrary B rather than typical one $\prod_x \delta[f[A(x)]] = B[f[A(x)]]$ which strictly imposes $f[A(x)]=0$.

achieved by taking $\xi = 0$.

while for non-zero ξ, $B[f[A(x)]]$ doesn't pick out specific gauge field that satisfies gauge fixing function, that is, gauge fixing becomes

$f[A(x)] = \omega(x)$, $\omega(x)$ is arbitrary fixed function of spacetime.



importantly, we still pick gauge field at each spacetime point